

Human single-neuron activity is modulated by intracranial theta burst stimulation of the basolateral amygdala

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eLife Assessment

This **important** study provides a description of how single-neuron firing rates in the human medial temporal lobe and frontal cortex are modulated by theta-burst stimulation of the basolateral amygdala. The results are supported by **convincing** evidence obtained from a rigorous task design and analysis of an incredibly rare dataset. The results may help guide future studies incorporating amygdala stimulation to improve patient health.

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Abstract

Direct electrical stimulation of the human brain has been used for numerous clinical and scientific applications. At present, however, little is known about how intracranial stimulation affects activity at the microscale. In this study, we recorded intracranial EEG data from a cohort of patients with medically refractory epilepsy as they completed a visual recognition memory task. During the memory task, brief trains of intracranial theta burst stimulation (TBS) were delivered to the basolateral amygdala (BLA). Using simultaneous microelectrode recordings, we isolated neurons in the hippocampus, amygdala, orbitofrontal cortex, and anterior cingulate cortex and tested whether stimulation enhanced or suppressed firing rates. Additionally, we characterized the properties of modulated neurons, clustered presumed excitatory and inhibitory neurons by waveform morphology, and examined the

extent to which modulation affected memory task performance. We observed a subset of neurons (~30%) whose firing rate was modulated by TBS, exhibiting highly heterogeneous responses with respect to onset latency, duration, and direction of effect. Notably, location and baseline activity predicted which neurons were most susceptible to modulation, although the impact of this neuronal modulation on memory remains unclear. These findings advance our limited understanding of how focal electrical fields influence neuronal firing at the single-cell level.

Highlights

- Individual neurons in the human brain were responsive to theta burst stimulation
- Basolateral amygdala stimulation preferentially modulated neurons in the hippocampus, orbitofrontal cortex, and amygdala
- Excitatory and inhibitory neurons were modulated equally
- Neurons modulated by stimulation tended to have greater baseline firing rates
- Duration, onset, and valence of neuronal modulation were heterogeneous

Introduction

The amygdala is a highly connected cluster of nuclei with input from multiple sensory modalities and vast projections to distributed cortical and subcortical regions involved in autonomic regulation and cognition (Kim et al., 2011 [↗](#); McDonald and Mott, 2017 [↗](#); Roy et al., 2009 [↗](#); Stein et al., 2007 [↗](#)). Numerous studies have described the amygdala's capacity to facilitate the encoding of long-lasting emotional memories (Dolcos et al., 2004 [↗](#); Hermans et al., 2014 [↗](#); LaBar and Cabeza, 2006 [↗](#); Phelps, 2006 [↗](#), 2004 [↗](#); Phelps and LeDoux, 2005 [↗](#); Qasim et al., 2023 [↗](#); Richardson et al., 2004 [↗](#); Roozendaal et al., 2009 [↗](#); Schwabe et al., 2013 [↗](#); Zheng et al., 2017 [↗](#)). Specific oscillatory dynamics in the amygdalohippocampal circuit are increasingly understood to be essential in prioritizing the encoding of these salient memories (Qasim et al., 2023 [↗](#); Zheng et al., 2019 [↗](#), 2017 [↗](#)).

Recently, direct electrical stimulation of the basolateral complex of the amygdala (BLA) in humans has revealed a more generalized ability to enhance declarative memory irrespective of the emotional valence (Inman et al., 2018 [↗](#)), likely by promoting synaptic plasticity-related processes underlying memory consolidation in the hippocampus and medial temporal lobe (McGaugh, 2013 [↗](#), 2004 [↗](#); McGaugh et al., 2002 [↗](#); Roesler et al., 2021 [↗](#)). These pro-memory effects were achieved with rhythmic theta-burst stimulation (TBS), which is known to induce long-term potentiation (LTP), a key mechanism in memory formation (Larson and Munkácsy, 2015 [↗](#)). Emerging evidence suggests that intracranial TBS may also enhance memory specificity (Titiz et al., 2017 [↗](#)), evoke theta-frequency oscillations (Solomon et al., 2021 [↗](#)), and facilitate short-term plasticity in local field potential recordings (Herrero et al., 2021 [↗](#); Huang et al., 2024 [↗](#)). However, the extent to which exogenous TBS modulates activity at the single-cell level and whether this modulation is associated with memory performance remains poorly understood.

Here, we address this knowledge gap by conducting simultaneous microelectrode recordings from prefrontal and medial temporal structures during a memory task in which intracranial TBS was applied to the BLA. To characterize neuronal modulation, we contrasted the trial-averaged, peri-stimulation firing rates across stim and no-stim conditions; neurons were further classified based

on their location, baseline activity, and direction of effect (enhancement vs. suppression). We predicted that intracranial TBS of the BLA would modulate spiking activity within highly connected regions (e.g., hippocampus) and improve memory task performance.

Methods

Participants

We report results from a cohort of 23 patients with medically refractory epilepsy who underwent stereoelectroencephalography to localize epileptogenic foci (74% female, 19–66 years old). All patients were age 18+ and able to provide informed consent. No exclusion was made concerning a patient's sex, gender, race, ethnicity, or socioeconomic status. Surgeries were performed at the University of Utah in Salt Lake City, UT (n = 10) and Barnes-Jewish Hospital in St. Louis, MO (n = 13). Patients were monitored continuously by a clinical team during their post-operative hospital course. Each patient signed a written informed consent form before participation in the research study; all study procedures were approved by the Institutional Review Board at the University of Utah (IRB 00144045, IRB 00114691) and Washington University (IRB 202104033).

Electrode Placement and Localization

Numbers and trajectories of stereoelectroencephalography electrode placements were determined case-by-case and solely derived from clinical considerations during a multidisciplinary case conference without reference to this research program. Each patient was implanted with clinical macroelectrodes and 1–3 Behnke-Fried depth electrodes (Ad-Tech Medical Instrument Corporation, Oak Creek, WI), which contained both macro- and microelectrode contacts (eight 40 μm diameter microwires and one unshielded reference wire) for recording local field potentials and extracellular action potentials, respectively (**Figure 1A**). To localize electrodes, we leveraged the open-access *Localize Electrodes Graphical User Interface (LeGUI)* (Davis et al., 2021) software developed by our group, which performs coregistration of pre-operative MRI and post-operative CT sequences, image processing, normalization to standard anatomical templates, and automated electrode detection.

Intracranial Electrophysiology

Neurophysiological data were recorded at both hospitals using a neural signal processor (Blackrock Microsystems, Salt Lake City, UT; Nihon Kodan USA, Irvine, CA) sampling at 30 kHz. Microelectrode contacts were locally referenced to a low-impedance microwire near the recording wires. Macroelectrode contacts were referenced to an intracranial contact located within the white matter with minimal activity, per recommended practices (Mercier et al., 2022).

Experimental Design

Patients completed a visual recognition memory task previously employed by our group to characterize the effects of basolateral amygdala stimulation upon memory consolidation (Inman et al., 2018). The memory task consisted of an encoding session, during which a series of neutral valence images were presented, and a self-paced retrieval session ~24 hours post-encoding wherein patients were asked to indicate whether each image onscreen was old (previously shown) or new (unseen) (**Figure 2A**). Data were aggregated across four highly similar experimental paradigms with minor differences in the content of visual stimuli, number of trials, stimulation parameters, and testing intervals. Each encoding session consisted of 160 or 320 trials wherein an image was presented on screen for 3 s, followed by a ~6 s interstimulus interval (fixation cross on screen).

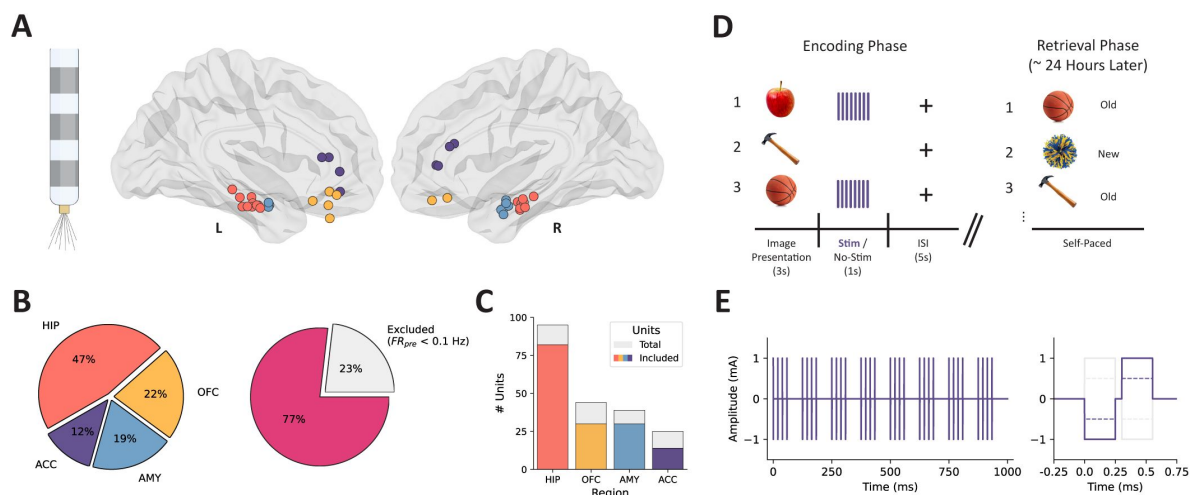


Figure 1

Microelectrode locations, unit counts, and experimental design.

(A) Behnke Fried-style macro/micro depth electrode (left) and microelectrode bundle locations projected in MNI space (right). (B) The proportion of units recorded from each brain area (left) and the proportion of units that met the criteria for inclusion in analyses (average pre-trial baseline firing rate ≥ 0.1 Hz) (right). (C) Counts of total (grey) and included (colored) units within each region. (D) Intracranial recording and stimulation took place in the context of a two-phase (encoding, retrieval) visual recognition memory task. A series of neutral valence images were shown (3 s), half of which were followed by direct electrical stimulation (1 s). Retrieval memory was tested during a self-paced task ~ 24 hours later. (E) Simulated theta-burst stimulation trace (left) and individual stimulation pulse (right); charge-balanced, bipolar, biphasic rectangular pulses were delivered over a 1 s period. HIP = hippocampus (coral), OFC = orbitofrontal cortex (yellow), AMY = amygdala (blue), ACC = anterior cingulate cortex (purple).

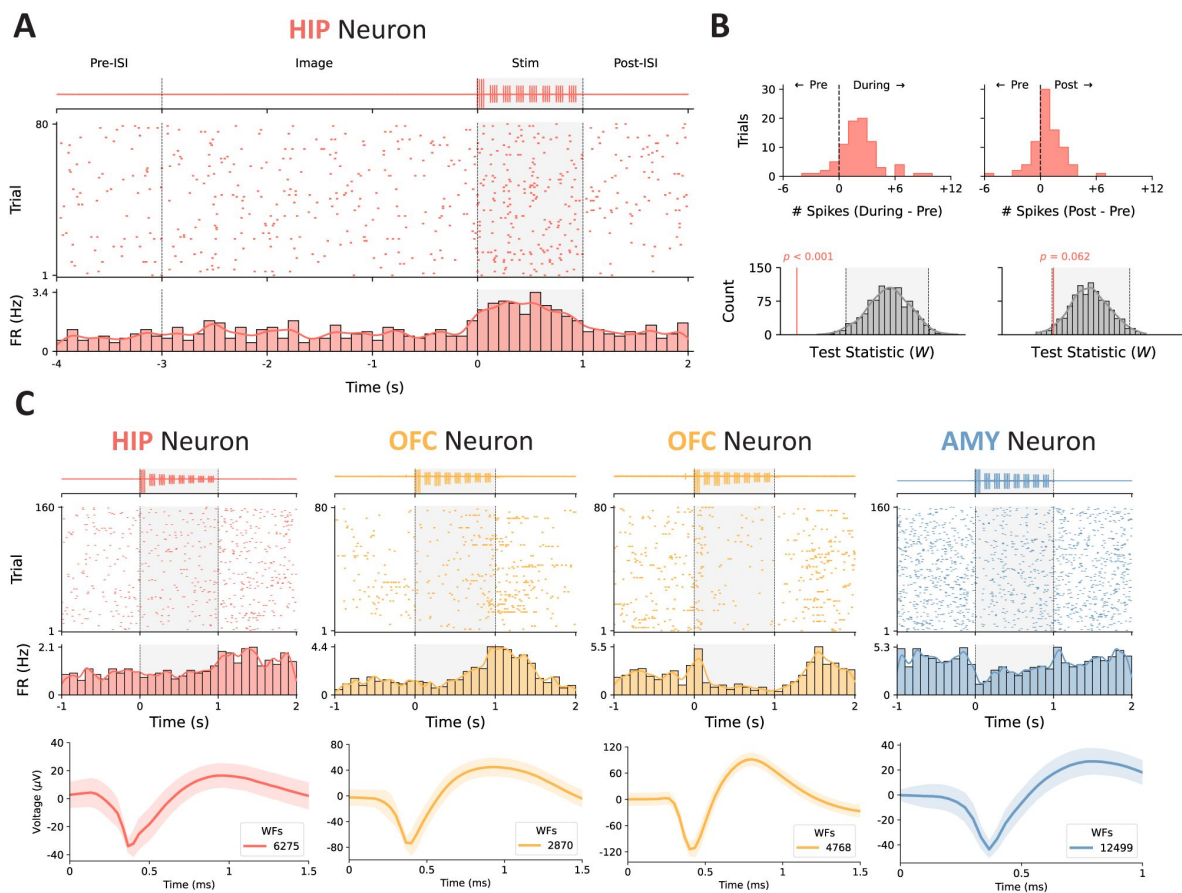


Figure 2

Example raster plots depicting heterogeneous responses to stimulation.

(A) Representative example of modulation during stimulation. The highpass-filtered, trial-averaged LFP from the corresponding microwire is shown (top) above the spike raster for an example unit located in the hippocampus (middle); the grey shaded region depicts the duration of stimulation with onset at $t = 0$. The average firing rate across trials was estimated by convolving the binned spike counts (100 ms bins) with a Gaussian kernel (bottom). (B) The difference in the number of spikes in the 1 s peri-stimulation epochs for each trial is shown (top). We subsequently performed a Wilcoxon signed-rank test on the during- and post-stimulation spike counts for each trial vs. the pre-trial baseline and compared the empirical test statistic against a null distribution generated by shuffling the epoch labels 1,000 times (bottom); the grey-shaded region represents the distribution containing 95% of observed values. (C) Some units (left, left-middle) exhibited increased firing rates, whereas others (right-middle, right) had their firing suppressed. The temporal dynamics of the firing rate modulation (e.g., onset, duration) were highly variable across units. The averaged waveform for each of the visualized units is shown below its corresponding peri-stimulation raster plot (WFs = waveforms); the shaded region represents standard deviation across waveforms.

Spike Detection and Sorting

Microelectrode data were first filtered between 250–500 Hz with a zero-phase lag bandpass filter and re-thresholded offline at -3.5 times the root mean square of the signal to identify spike waveforms. Units were isolated during a semi-automated process within *Offline Sorter* (Plexon Inc, Dallas, TX) by applying the T-distribution expectation maximization method on the first three principal components of the detected waveforms (initial parameters: degrees of freedom multiplier = 4, initial number of units = 3) (Shoham et al., 2003 [↗](#)). Finally, the waveform shapes, interspike interval distribution, consistency of firing, and isolation from other waveform clusters were manually inspected for further curation and removal of spurious, non-physiological threshold crossings that could represent stimulation artifacts.

Single Unit Quality Metrics

We calculated several distinct metrics to characterize detected units' properties and assess the quality of our spike sorting (*Figure S2*): the number of units detected per microelectrode bundle, the mean firing rate (Hz) for each unit, the percentage of interspike intervals < 3 ms, the coefficient of variation across each unit's spike train, the average presence ratio of firing in 1 s bins (proportion of bins which contain ≥ 1 spike), the ratio between the peak amplitude of the averaged waveform and its standard deviation, and the mean signal-to-noise ratio of the averaged waveform.

Discrimination of Excitatory vs. Inhibitory Neurons

We calculated two metrics from the averaged waveform from each detected unit: the valley-to-peak-width (VP) and the peak half-width (PHW) (**Figure 4A** [↗](#)); previously, these two properties of waveform morphology have been used to discriminate pyramidal cells (excitatory) from interneurons (inhibitory) in human intracranial recordings (Peyrache et al., 2012 [↗](#)). Next, we performed k-means clustering ($n = 2$ clusters) on the waveform metrics, in line with previous approaches to cell type classification.

Intracranial Theta Burst Stimulation

We delivered direct electrical stimulation to the basolateral amygdala during half of the trials in the encoding phase of each experimental session. Stimulation pulses were delivered immediately once the image was removed from the screen and in a patterned rhythm designed to entrain endogenous theta-gamma oscillatory interactions (i.e., theta-burst stimulation, TBS) (Hanslmayr et al., 2016 [↗](#); Suppa et al., 2016 [↗](#)). Specifically, we administered current-controlled, charge-balanced, bipolar, 1 mA, biphasic rectangular pulses over a 1 s period with a 50% duty cycle. Stimulation pulses were delivered at a rate of 50 Hz and nested within eight equally-spaced bursts (~ 8 Hz) (**Figure 2B** [↗](#)–C). A subset of experiments ($n = 7$) used a lower current (0.5 mA) with variable pulse frequencies across trials (33 Hz, 50 Hz, 80 Hz).

Peri-Stimulation Modulation Analyses

We first created peri-stimulation epochs (1 s pre-trial ISI, 1 s after image onset, 1 s during stimulation/after image offset, 1 s post-stimulation), with $t = 0$ representing stimulation onset and the moment at which the image was removed from the screen (**Figure 2A** [↗](#)); identical epochs were created for the image-only (no-stimulation) trials. Each epoch was constrained to 1 s to ensure that subsequent firing rate contrasts were unbiased and to capture potential transient effects (e.g., image onset/offset). Units with a trial-averaged baseline (pre-trial ISI) firing rate of < 0.1 Hz were excluded from subsequent analyses because low firing rates limited the ability to detect modulation robustly. Units were designated as “modulated” if either the during- or post-

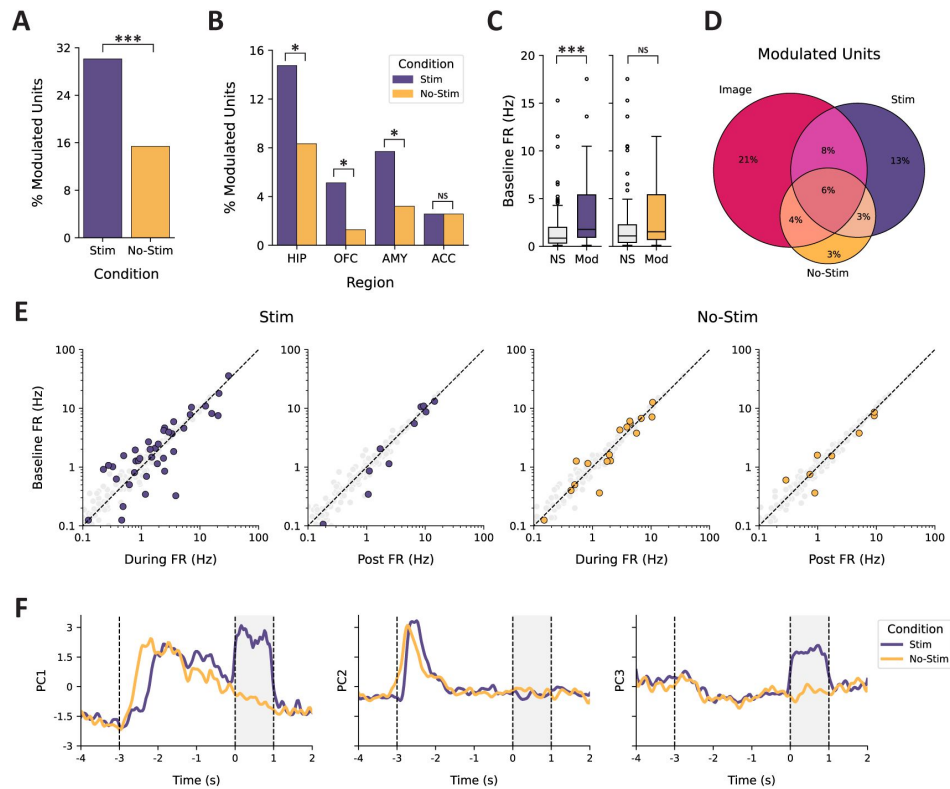


Figure 3

Characterization of modulation in neuronal firing rate.

(A) Percent of modulated units observed across trials separated by stim (purple) vs. no-stim conditions (orange). (B) Percent of modulated units as a function of recording region. (C) Comparison of baseline firing rate in units separated by condition (stim vs. no-stim) and outcome (NS = not significant, Mod = modulated). (D) Venn diagram depicting the shared and independent proportions of units modulated by image onset (Image) and the two experimental conditions (stim vs. no-stim). (E) Scatterplot of pre-stimulation firing rate relative to the firing rate during the two contrast windows (during, post) for the stim (left) and no-stim (right) conditions. Modulated units are highlighted in purple (stim) or orange (no-stim), whereas units without a significant change are shown in grey. (F) Temporal dynamics of pseudo-population coactivity within each condition, represented by the first three principal components of the trial-averaged firing rates. The grey-shaded region depicts the duration of stimulation with onset at $t = 0$. Images were presented on screen for 3 s, with onset at $t = -3$. * $p < 0.05$, *** $p < 0.001$, NS = not significant.

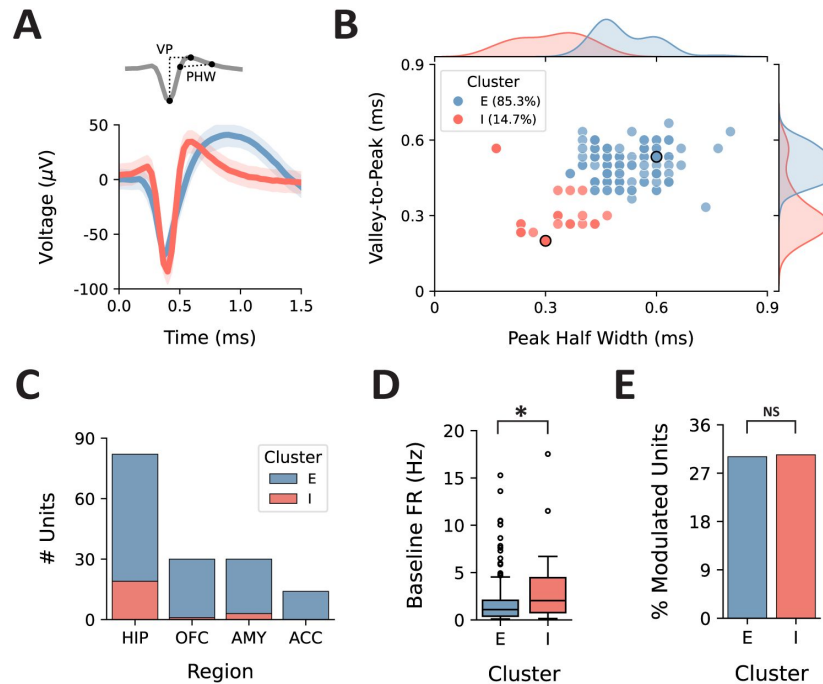


Figure 4

Separation of presumed excitatory and inhibitory neurons by waveform morphology

(A) Two metrics were calculated using the averaged waveforms for each detected unit: the valley-to-peak width (VP) and peak half-width (PHW). (B) Scatterplot of the relationship between VP and PHW; note that units with identical metrics are overlaid. Using k-means clustering, we identified two distinct response clusters, representing presumed excitatory (E, blue) and inhibitory (I, red) neurons. The units from which the example waveforms were taken are outlined in black. Probability distributions for each metric are shown along the axes. (C) Total number of units within each cluster, separated by region. (D) Comparison of baseline firing rates, separated by cluster. (E) Percent of modulated units in each cluster. * $p < 0.05$, NS = not significant.

stimulation firing rate contrast was significant following permutation testing (described in *Statistical Analyses*). An additional contrast of pre-trial ISI vs. image onset was performed to evaluate the sensitivity of neurons to task stimuli (i.e., image presentation).

To investigate whether stimulation altered rhythmicity in neuronal firing, we analyzed the spike timing autocorrelograms. More specifically, we computed the pairwise differences in spike timing for each trial (bin size = 5 ms, max lag = 250 ms) and then contrasted the differences in the latencies of the peak normalized autocorrelation value between epochs (pre-, during-, post-stimulation). Only neurons with a firing rate of ≥ 1 Hz ($n = 70/203$, 34.5%) were included in this analysis since sparse firing resulted in noisy autocorrelation estimates.

Finally, we measured modulation in multiunit activity (MUA) by filtering the microelectrode signals in a 300-3,000 Hz window and counting the number of threshold crossings. Thresholds were determined on a per-channel basis and defined as -3.5 times the root mean square of the signal during the baseline period; activity during stimulation was excluded since stimulation artifact is difficult to separate from MUA in the absence of spike sorting.

Population Analyses

To analyze pseudo-population activity, we performed a linear dimensionality reduction with PCA on a matrix of the z-scored trial-averaged firing rates of all neurons recorded across patients (*sklearn.decomposition.PCA*). This approach enables the identification of dominant patterns of coordinated neural activity that may not be apparent when examining individual neurons in isolation. Doing so allowed us to qualitatively examine the temporal dynamics of the dominant modes of neuron coactivity in a low-dimensional subspace, separated by experimental condition (stim vs. no-stim), region, and laterality relative to stimulation. The firing rate matrices for each condition were concatenated prior to PCA to facilitate direct comparison among the principal components; for simplicity, only the first three principal components are visualized. By collapsing across subjects into a common pseudo-population, this analysis provides a mesoscale view of how stimulation modulates shared activity patterns across anatomically distributed neural populations.

Statistical Approach

All statistical analyses were conducted using custom Python scripts and established statistical libraries (i.e., Scikit-learn (Pedregosa et al., 2012 [↗](#)), *Scipy* (Virtanen et al., 2020 [↗](#)), *Statsmodels* (Seabold and Perktold, 2010 [↗](#))). We performed two separate Wilcoxon signed-rank tests across trials on the during- and post-stimulation spike counts relative to their corresponding pre-trial baseline spike counts. To control for false positives, we compared the empirical test statistic against a null distribution generated from shuffling pre/during/post epoch labels ($n = 1000$ permutations) (**Figure 2B** [↗](#)). An identical analysis was also performed on the no-stimulation (image-only) trials.

To test for differences in the proportion of modulated units (across conditions, regions, cell type, stimulation parameters, and behavioral outcomes), we performed a series of one- and two-sided Fisher's exact tests. Next, we used Mann-Whitney U tests to contrast baseline firing rates among modulated vs. unaffected units. Behavioral performance during the memory task was calculated using d' , defined as the difference in an individual's z-scored hit rate and false alarm rate. Observed changes in recognition memory were split into two categories using a d' difference threshold of ± 0.5 : responder ($\Delta d' < -0.5$ or $\Delta d' > +0.5$) or non-responder ($-0.5 \leq \Delta d' \leq 0.5$). The threshold of ± 0.5 was chosen based on the defined range of a "medium effect" for Cohen's d , which bears conceptual similarity to d' . To test the hypothesis that stimulation affected behavioral performance, we used a linear mixed effects model with d' score as the dependent variable,

condition and experiment as fixed effects, and session as a random effect; an additional test for differences among hit rates (percent of previously seen images correctly identified) was implemented using a paired-samples t-test.

Firing Rate Control Analyses

We performed a series of control analyses to test whether our approach to firing rate detection was robust. First, we performed a sensitivity analysis by systematically varying the baseline firing rate threshold used to exclude units from modulation analyses. The threshold for inclusion of units was varied from 0–3 Hz (0.1 Hz step size), and the firing rate analyses were repeated to quantify the proportion of units meeting inclusion criteria and the proportion of units designated as modulated (*Figure S7A*). Next, we performed a dropout analysis wherein segments of data near the onset of a stimulation burst were removed from the during-stimulation epoch (an identical segment was also removed from the pre-trial ISI and post-stimulation epochs). To this end, we removed a window of data starting at the onset of each burst spanning 0–60 ms (5 ms step size, eight bursts in train) and recomputed the proportion of units meeting inclusion criteria and the proportion of units designated as modulated (*Figure S7B*). Finally, to better understand the tradeoffs with our statistical approach, we generated synthetic data with different baseline firing rates (0.1–5.0 Hz) and effect sizes (± 0.1 –0.7 Hz), then simulated the likelihood of detecting modulation across variable conditions (*Figure S7C*).

Results

We recorded single-unit activity from 23 patients ($n = 30$ sessions) with medically refractory epilepsy as they completed a visual recognition memory task (see *Table S1* for patient demographics). During the encoding session of the experiment, each patient received either 80 or 160 trials of bipolar intracranial TBS to a contiguous pair of macroelectrode contacts in the BLA (see *Figure S1* for anatomical localization of stimulated contacts. An equal number of “no-stimulation” trials were randomly interspersed to evaluate the effect of stimulation on memory performance and control for neuronal modulation resulting from experimental stimuli (e.g., image presentation). In total, we isolated 203 putative neurons from 68 bundles of 8 microwires each, distributed among recording sites in the hippocampus (HIP, $n = 95$ units), orbitofrontal cortex (OFC, $n = 44$), amygdala (AMY, $n = 39$), and anterior cingulate cortex (ACC, $n = 25$) (**Figure 1**; see also *Figure S2* and *Figure S3*); a subset of these units ($n = 47$, 23.2%) was excluded from subsequent analyses because low baseline firing rates (< 0.1 Hz) limited the ability to robustly detect modulation.

Theta Burst Stimulation of the BLA Modulates Widely Distributed Populations of Neurons

We hypothesized that BLA stimulation would modulate neuronal activity in the sampled regions, given the amygdala’s well-established connectivity to the HIP, OFC, and ACC (Roy et al., 2009). To test this hypothesis, we quantified spike counts across trials within peri-stimulation epochs (1 s pre-trial ISI, 1 s after image onset, 1 s during stimulation/after image offset, and 1 s post-stimulation) and used Wilcoxon signed-rank tests to compare the spike counts against a null distribution generated by shuffling epoch labels. We performed two firing rate contrasts across trials (pre-trial ISI vs. during stimulation, pre-trial ISI vs. post-stimulation) within two distinct conditions (stim, no-stim); an additional contrast of the pre-trial ISI vs. image onset epochs was included to evaluate the sensitivity of neurons to task image presentations (**Figure 2A**).

BLA TBS modulated firing rates in 30.1% of all recorded units, a significantly higher proportion than the 15.4% responsive to no-stim (image only) trials (one-sided Fisher’s exact test, $OR = 2.37$, $p < 0.001$; **Figure 3A**). Across all regions sampled, we observed units modulated by the stim and

no-stim conditions. Units in HIP (OR = 2.07, $p = 0.044$), OFC (OR = 5.09, $p = 0.040$), and AMY (OR = 3.33, $p = 0.042$) were most sensitive to stimulation; we did not observe a difference in the proportions of units within the ACC responsive to the stim vs. no-stim conditions (OR = 1.00, $p = 0.661$; **Figure 3B**). Only 9.0% of units responded to both the stim and no-stim conditions, despite approximately half of the stim-modulated units (representing 14.7% of all units) exhibiting a change in firing rate associated with image onset (**Figure 3D**). This result suggests that the units modulated by stimulation are largely distinct from those responsive to image offset during trials without stimulation. Stimulation, however, did not appear to alter the rhythmicity in neuronal firing, as measured by spiking autocorrelograms (*Figure S5*).

Because neuronal firing properties vary across cell types (Barthó et al., 2004; Keller et al., 2010; Peyrache et al., 2012; Quyen et al., 2008), we also tested whether baseline (pre-trial ISI) firing rates predicted a unit's response to stimulation, suggestive of selective engagement of specific neuron populations. Stimulation-modulated units exhibited significantly higher baseline firing rates compared to unaffected units ($U(N_{\text{Stim, Mod}} = 47, N_{\text{Stim, NS}} = 109) = 3450.50, p < 0.001$). No difference in baseline firing rate was observed among units modulated in the no-stim condition, compared to those that were unaffected ($U(N_{\text{No-Stim, Mod}} = 24, N_{\text{No-Stim, NS}} = 132) = 1964.00, p = 0.062$) (**Figure 3C**). The median (Q1, Q3) baseline firing rates for modulated units in the stim and no-stim conditions were 1.77 Hz (0.95 Hz, 5.39 Hz) and 1.53 Hz (0.72 Hz, 5.41 Hz), respectively.

In a subset of experimental sessions ($n = 7$), we explored the effects of different stimulation parameters on neuronal modulation within an experimental session; more specifically, we employed a lower stimulation amplitude (0.5 mA vs. 1.0 mA) and varied pulse frequency (33 Hz vs. 50 Hz and 33 Hz vs. 80 Hz). Neither the amplitude of stimulation (OR = 1.69, $p = 0.302$, $n = 30$) nor pulse frequency (33 vs. 80 Hz; OR = 0.00, $p = 1.000$, $n = 1$; 50 vs. 80 Hz; OR = 1.40, $p = 0.758$, $n = 6$) significantly altered the proportion of modulated units. (*Figure S4B*).

Our exploratory analyses of pseudo-population activity revealed interesting temporal dynamics associated with image presentation and the delivery of stimulation. More specifically, we observed variation among the first three principal components across both stim and no-stim trials associated with image presentation ($t = -3$ to $t = 0$) and robust shifts in coactivity modes associated with stimulation ($t = 0$ to $t = 1$) (**Figure 3F**). Further characterization of these dynamics suggests that TBS of the BLA primarily resulted in transient changes to firing rate coactivity within hippocampal neurons and was present regardless of the neuron's laterality to stimulation (*Figure S6A–B*). Taken together, these analyses reveal global structure in the state space of responses to BLA stimulation within hippocampal circuits.

Finally, we performed three supplementary analyses to evaluate the robustness of our approach to detecting firing rate modulation: a sensitivity analysis assessing the proportion of modulated units at different firing rate thresholds for inclusion/exclusion, a data dropout analysis designed to control for the possibility that non-physiological stimulation artifacts may preclude the detection of temporally adjacent spiking, and a synthetic detection probability analysis. These results recapitulate our observation that units with higher baseline firing are most likely to exhibit modulation (though the probability of detecting modulation is lower for sparsely active neurons) and suggest that suppression in firing rate is not solely attributable to amplifier saturation following stimulation (*Figure S7*).

Neurons Exhibit Heterogenous Responses to Theta Burst Stimulation

Recent studies have reported enhanced neural plasticity (via intracranial local field potential recordings and evoked responses) following repetitive direct electrical stimulation (Herrero et al., 2021; Huang et al., 2024, 2019; Keller et al., 2018). Accordingly, we hypothesized that

recorded units would predominantly exhibit enhanced spiking in response to intracranial TBS of the BLA. Similarly, individual units exhibited highly variable responses to stimulation with respect to onset latency (rapid vs. delayed), duration (transient vs. durable), and valence (enhancement vs. suppression) (**Figure 2B** [↗](#)).

The most common epoch for firing rate modulation was during the 1 s epoch in which TBS was delivered (25.0% of all neurons). Smaller subsets were modulated only in the 1 s post-stimulation epoch (6.4%) or in both the during- and post-stimulation epochs (1.3%). A similar trend was observed for modulation in the no-stim condition: 10.9% during, 5.8% post, and 1.3% for both. Suppression was most common among modulated units during stimulation (56.4%), whereas enhancement was the dominant response post-stimulation (70.0%). In contrast, enhancement was most common within both epochs across no-stim trials (58.5% during, 66.7% post). The mean ($\pm$ SD) absolute z-scored difference in firing rate across stimulation trials (relative to pre-trial ISI) was $z = 0.60 (\pm 0.58)$ and $z = 0.43 (\pm 0.27)$ for the during- and post-stimulation epochs, respectively. Across no-stim trials, we observed a mean absolute z-scored difference of $z = 0.38 (\pm 0.24)$ and $z = 0.30 (\pm 0.18)$ in analogous epochs (**Figure 3E** [↗](#)). Additional characterization of MUA revealed a dominant signature of increased activity post-vs. pre-stimulation, in line with these trends observed at the single-neuron level (*Figure S8*).

Theta Burst Stimulation Modulates Excitatory and Inhibitory Neurons Equally

Using k-means clustering, we grouped neurons into two distinct clusters based on waveform morphology, representing neurons that were presumed to be excitatory (E) and inhibitory (I) (**Figure 4B** [↗](#)). Inhibitory (fast-spiking) neurons exhibited shorter waveform VP and PHW, compared with excitatory (regular-spiking) neurons (I cluster centroid: VP = 0.50ms, PHW = 0.51ms; E cluster centroid: VP = 0.32ms, PHW = 0.31ms), and greater baseline firing rates ($U(N_I = 23, N_E = 133) = 1074.50, p = 0.023$) (**Figure 4D** [↗](#)). Although we observed a much greater proportion of excitatory vs. inhibitory neurons (E: 85.3%, I: 14.7%), stimulation appeared to affect excitatory and inhibitory neurons equally, suggesting that one cell type is not preferentially activated over another (**Figure 4E** [↗](#)).

Association Between Neuronal Modulation and Memory Performance is Unclear

Next, we investigated the link between stimulation and performance during the visual recognition memory task. To this end, we first used a linear mixed-effects model to examine the effect of condition (stim, no-stim) on memory performance (d') across trials in each session, with individual sessions treated as a random effect (intercept). Experiment type was also included as a fixed effect since data were aggregated across four highly similar experiments with minor differences in the content of visual stimuli, number of trials, stimulation parameters, and testing intervals. We did not observe an overall effect of memory enhancement ($p > 0.05$) when controlling for subject-level variability (**Figure 5A** [↗](#)). The lack of a memory enhancement effect may be associated with high hit rates limiting sensitivity (mean $\pm$ SD) ($75.7\% \pm 13.5\%$ for no-stim trials, $75.0\% \pm 14.3\%$ for stim trials), and considerable variability among false alarm rates across participants ($17.9\% \pm 17.4\%$, range 0–70%; **Figure 5B** [↗](#)).

At the level of individual sessions, we observed enhanced memory ($\Delta d' > +0.5$) in 36.7%, impaired memory ($\Delta d' < -0.5$) in 20.0%, and negligible change ($-0.5 \leq \Delta d' \leq 0.5$) in 43.3% when comparing performance between the stim and no-stim conditions; a threshold of $\Delta d' \pm 0.5$ was chosen for this classification based on the defined range of a “medium effect” for Cohen’s d . To test our hypothesis that neuronal modulation would be associated with changes in memory performance, we combined the sessions that resulted in either memory enhancement or impairment and

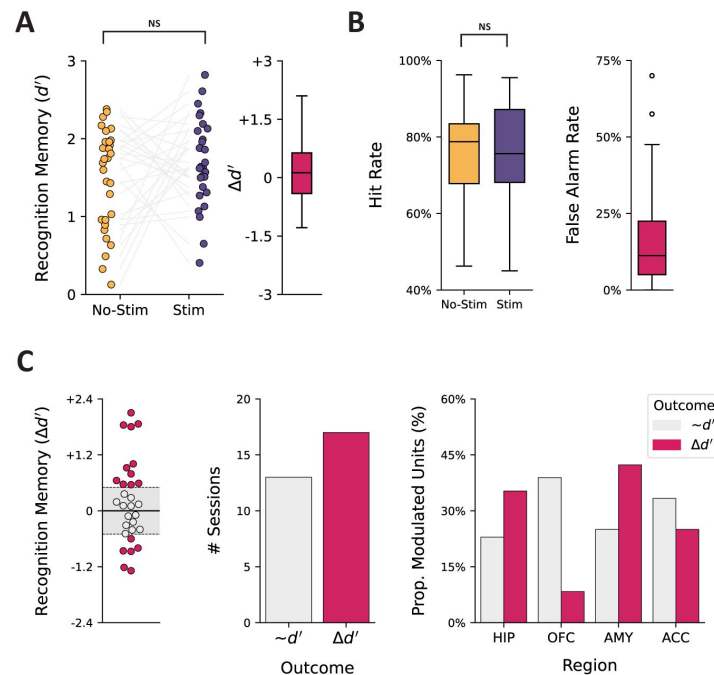


Figure 5

Summary of behavioral performance during memory task.

(A) Memory performance for each session is quantified using d' (left); grey lines connect d' scores across conditions for an individual session. Boxplot of the observed difference in d' scores across conditions (right). (B) Hit rate (percent of old images correctly recognized) and false alarm rate (percent of new images incorrectly labeled as old) across conditions. (C) Change in recognition memory performance was split into two categories using a d' difference threshold of ± 0.5 : responder (positive or negative; $\Delta d'$, pink) and non-responder ($\sim d'$, grey). Individual d' scores are shown (left) with points colored by outcome category; dotted lines demarcate category boundaries, and the grey-shaded region represents negligible change. The number of sessions within each outcome category (middle) and the proportion of modulated units as a function of outcome category, separated by region (right). NS = not significant.

contrasted the proportion of modulated units across regions sampled. We did not, however, observe a meaningful difference in the proportion of modulated units when grouped by behavioral outcome (all contrasts $p > 0.05$) (**Figure 5C** [↗](#)).

Proximity to white matter has been shown to influence the effects of stimulation on behavior and the strength of evoked responses ([Mankin et al., 2021](#) [↗](#); [Mohan et al., 2020](#) [↗](#); [Paulk et al., 2022](#) [↗](#)). Across all stimulated contacts, we observed only small differences in the proximity of stimulation contacts to white matter (median = 4.6 mm, range = 1.5–8.0 mm), likely because the chosen target (i.e., basolateral amygdala) has several nearby white matter structures (e.g., stria terminalis). Nonetheless, we performed a linear regression between the proximity to white matter and the stimulation-induced effect on behavior (stimulation vs. no-stimulation d' difference), the results of which indicate no clear association ($p > 0.05$; see *Figure S9*).

Discussion

Theta-burst stimulation is an efficient and validated paradigm for inducing long-term potentiation (LTP) in neural circuits ([Larson and Munkácsy, 2015](#) [↗](#)). Additionally, intracranial TBS was recently shown to promote region-specific short-term plasticity ([Herrero et al., 2021](#) [↗](#); [Huang et al., 2024](#) [↗](#)) and entrain frequency-matched oscillations ([Solomon et al., 2021](#) [↗](#)). At present, however, there is an incomplete understanding of how these population-level responses to stimulation relate to a modulation in the activity of individual neurons, which are thought to be the substrate of memory encoding and retrieval ([Hebb, 1949](#) [↗](#)). Here, we address this knowledge gap by characterizing neuronal firing recorded from microelectrodes in humans undergoing intracranial TBS of the BLA. Our experimental design focused explicitly on stimulation of the BLA, given our prior work that seeks to understand amygdala-mediated memory enhancement in humans ([Inman et al., 2023](#) [↗](#), [2020](#) [↗](#), [2018](#) [↗](#)).

We observed neurons distributed throughout the hippocampus, orbitofrontal cortex, anterior cingulate cortex, and amygdala responsive to direct electrical TBS. The effect of TBS on firing rate was heterogeneous with respect to onset, duration, and valence. Previous work characterizing local-field potential responses to intracranial TBS observed similarly bidirectional effects throughout the brain suggestive of short-term plasticity ([Huang et al., 2024](#) [↗](#)). Few studies, however, have characterized the impact of exogenous stimulation on the spiking activity of individual neurons, none of which involve intracranial theta burst stimulation. One study reported a long-lasting reduction in neural excitability among parietal neurons, with variable onset time and recovery following continuous transcranial TBS in non-human primates ([Romero et al., 2022](#) [↗](#)). In a similar vein, it was recently shown that human neurons are largely suppressed by single-pulse electrical stimulation ([Cowan et al., 2024](#) [↗](#); [Van der Plas et al., 2024](#) [↗](#)). Other emerging evidence suggests that transcranial direct current stimulation may entrain the rhythm rather than rate of neuronal spiking ([Krause et al., 2019](#) [↗](#)) and that stimulation-evoked modulation of spiking may meaningfully impact behavioral performance on cognitive tasks ([Fehring et al., 2024](#) [↗](#)). An alternative approach has focused on the delivery of spatially selective microstimulation resembling the extracellular currents that normally modulate neuronal activity—this methodology has been used to bi-directionally drive neuronal firing in human temporal cortex ([Youssef et al., 2023](#) [↗](#)) and enhance memory specificity for images following stimulation ([Titiz et al., 2017](#) [↗](#)).

Although subsets of neurons from each region we sampled were responsive to stimulation, we observed the greatest difference in the proportion of modulated units across conditions in the hippocampus, orbitofrontal cortex, and amygdala. This regional selectivity is to be expected, given that numerous studies have characterized how structural, functional, and effective connectivity among brain regions predicts the effects of stimulation ([Fox et al., 2020](#) [↗](#); [Huang et al., 2024](#) [↗](#), [2019](#) [↗](#); [Keller et al., 2018](#) [↗](#), [2011](#) [↗](#); [Solomon et al., 2018](#) [↗](#); [Stiso et al., 2019](#) [↗](#)). We also

observed that units with greater baseline activity were most likely to exhibit modulated firing rates following stimulation. Other studies have identified firing patterns and waveform properties that differ between inhibitory and excitatory neurons in humans (Barthó et al., 2004 [↗](#); Keller et al., 2010 [↗](#); Peyrache et al., 2012 [↗](#); Quyen et al., 2008 [↗](#)). For example, baseline firing rate disambiguates “regular-spiking” and “fast-spiking” neurons, which are presumed to represent pyramidal cells and interneurons, respectively. To test these hypotheses directly, we clustered neurons into presumed excitatory and inhibitory neurons based on waveform morphology. In doing so, we observed ~85% excitatory and ~15% inhibitory neurons, which is very similar to what has been reported previously in human intracranial recordings (Cowan et al., 2024 [↗](#); Peyrache et al., 2012 [↗](#)). Interestingly, stimulation appeared to modulate approximately the same proportion of neurons for each cell type (~30%), despite the differently-sized groups. Recent reports, however, have suggested that the extent to which electrical fields entrain neuronal spiking, particularly with respect to phase-locking, may be specific to distinct classes of cells (Lee et al., 2024 [↗](#)).

Modulation in neuronal activity was defined by contrasting firing rates before, during, and after TBS across trials. In doing so, we were able to characterize coarse differences in activity indicative of enhancement or suppression. This approach, however, did not allow for analysis of more subtle, nuanced effects such as entrainment of spiking to individual bursts or pulses of TBS. Characterizations of rhythmicity in firing were challenging, given that most of the neurons we identified exhibit sparse activity with low baseline firing rates, and stimulation often resulted in further suppression of spiking.

Although stimulation artifacts resulted in amplitude threshold crossings that may be spuriously interpreted as a neuronal spike, we implemented several methods to mitigate the influence of non-physiological activity. First, the characteristics of each unit (e.g., waveform shape) were manually inspected during spike sorting and further quantified using several quality control metrics (e.g., interspike intervals); stimulation resulted in a stereotyped response that was easily detectable and removed from subsequent analyses. Additionally, we tested for modulation both during stimulation and in the post-stimulation epochs—a period in which no artifact was present. Contrary to what would be expected if stimulation artifacts were falsely elevating firing rates, we observed predominantly suppression during stimulation and enhancement post-stimulation.

Since we collected our microelectrode recordings in the context of a visual recognition memory task, we tested whether stimulation resulted in a change in memory performance. Although we hypothesized that stimulation would improve memory performance, we did not identify an apparent stimulation-related memory enhancement when controlling for individual differences, in contrast to our prior work (Inman et al., 2018 [↗](#)). The absence of such an effect may be, at least in part, attributable to the considerable variability that we observed in this cohort; indeed, baseline memory performance and individual differences have been shown to account for a substantial portion of the variability in this amygdala-mediated memory enhancement effect (Hollern et al., 2024 [↗](#)).

Several studies on rats have demonstrated that brief electrical stimulation of the BLA can prioritize the consolidation of specific memories (Bass et al., 2014 [↗](#), 2012 [↗](#); Bass and Manns, 2015 [↗](#); Manns and Bass, 2016 [↗](#)). These pro-memory effects emerged ~24 hours post-encoding and appear to be hippocampal-dependent (Bass et al., 2014 [↗](#)), despite not resulting in a net change in the firing rates of hippocampal pyramidal neurons; instead, BLA stimulation resulted in brief periods of spike-field and field-field synchrony within CA3–CA1 in the low-gamma frequency range (30–55 Hz), which may facilitate spike-timing-dependent plasticity in recently active neurons (Bass and Manns, 2015 [↗](#)).

The present study did not investigate interactions between spiking activity and local field potentials because neuronal spiking was sparse at baseline and often further suppressed by stimulation; only a very small proportion of the total number of trials across all neurons exhibited ≥ 10 spikes in both the 1 s pre- and post-stimulation epochs ($\sim 2.5\%$). Although certain metrics are not biased by sample size (e.g., pairwise phase consistency), low spike counts can dramatically affect variance and, therefore, result in unstable estimates (Vinck et al., 2011 [↗](#)).

How exactly the activity of single neurons is aggregated to produce local field potentials, which in turn interact with neuronal ensembles distributed throughout the brain, remains an active area of research (Herreras, 2016 [↗](#); Kajikawa and Schroeder, 2011 [↗](#); Manning et al., 2009 [↗](#); Teleńczuk et al., 2020 [↗](#)). One recent study that leveraged closed-loop stimulation targeting memory consolidation during sleep observed neuronal spiking with greater phase-locking to medial temporal lobe slow-wave activity following stimulation (Geva-Sagiv et al., 2023 [↗](#)); neuronal phase-locking, particularly to hippocampal theta oscillations, has long been associated with robust memory encoding and retrieval (Jacobs et al., 2007 [↗](#); Rutishauser et al., 2010 [↗](#); Schonhaut et al., 2024 [↗](#); Yoo et al., 2021 [↗](#)). Further characterization of these spike-field interactions and refinement of closed-loop stimulation methods may provide a means for precisely modulating neuronal dynamics, for example, by entraining neuronal spiking that is phase-aligned to endogenous hippocampal theta oscillations to selectively enhance the encoding or retrieval of memories (Hasselmo, 2005 [↗](#); Hasselmo et al., 2002 [↗](#); Hasselmo and Stern, 2014 [↗](#); Siegle and Wilson, 2014 [↗](#)).

Finally, we performed an exploratory analysis of neuronal pseudo-population activity, which suggests that hippocampal neurons exhibit robust changes in firing rate coactivity in response to BLA stimulation. Related research has similarly described how BLA stimulation can induce synchronous firing of hippocampal neurons, which has memory-enhancing effects (Bass and Manns, 2015 [↗](#)). Other studies have leveraged similar, low-dimensional representations of population dynamics to describe how coordinated neural activity facilitates inferential reasoning and memory retrieval within the medial frontal cortex and hippocampus, respectively (Courellis et al., 2024 [↗](#); Minxha et al., 2020 [↗](#)). Thus, a greater understanding of how neuronal coactivity may be precisely modulated by stimulation may help to refine therapeutic interventions targeting complex cognition and computation.

Conclusions

By characterizing patterns of neuronal modulation evoked by intracranial TBS, we provide new insights that link micro- and macroscale responses to stimulation of the human brain. These insights advance our limited understanding of how focal electrical fields influence neuronal firing at the single-cell level and motivate future neuromodulatory therapies that aim to recapitulate specific patterns of activity implicated in cognition and memory.

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Additional information

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Resource Availability

Custom Python analysis scripts used in the manuscript are publicly available on GitHub (<https://github.com/Justin-Campbell/BLAESUnits>). Deidentified neural recordings may be made available upon reasonable request.

Additional files

Figure S1. Anatomical location of stimulated electrodes. A coronal slice from the T1-weighted MRI scan is shown for each patient who participated in the study ($n = 16$). Electrode contacts within the same plane of the image are shown with blue circles, and the bipolar pair of stimulated contacts within the basolateral amygdala is highlighted in red. [↗](#)

Figure S2. Unit quality metrics. (A) Number of units detected per implanted microelectrode bundle. (B) Mean firing rate (Hz) across recording session. (C) Percent of interspike intervals < 3 ms. (D) Interspike interval coefficient of variation. (E) Mean presence ratio of firing within units (1 s bins). (F) Signal-to-noise ratio of unit waveform peak. (G) Mean signal-to-noise ratio across the entire unit waveform. (H) Representative example of stereotyped, high-amplitude stimulation-artifact waveform; non-physiological waveforms were excluded from analysis. [↗](#)

Figure S3. Characterization of units based on laterality relative to stimulation. (A) Unit counts on the contralateral (Contra) or ipsilateral (Ipsi) side of stimulation. (B) Unit counts separated by laterality and region. (C) Stacked histograms of Euclidean distance between microelectrode bundle location and stimulation contacts, separated by laterality (bin size = 5 mm). [↗](#)

Figure S4. Sub-analysis of stimulation parameters used across experiments. Comparison of the proportion of stimulation-modulated units across sessions with 1 mA ($n = 23$) vs. 0.5 mA ($n = 7$). (B) Comparison of the proportion of stimulation-modulated units across sessions testing distinct pulse frequencies: 33 Hz vs. 80 Hz ($n = 1$) and 50 Hz vs. 80 Hz ($n = 6$). The values above individual bars represent the number of sessions using that stimulation parameter. NS = not significant. [↗](#)

Figure S5. Analysis of modulation in spiking rhythmicity. (A) Representative autocorrelograms ACG for a single neuron. The pairwise differences in spike timing were computed for each trial and epoch (bin size = 5 ms, max lag = 250 ms), then smoothed with a Gaussian kernel. The peak in the normalized ACG across trials was computed for each epoch. (B) Kernel density estimate of the peak ACG lag, separated by epoch. (C) The peak ACG lags were split by whether the neuron was modulated (Mod) or unaffected by stimulation (NS = not significant) for each of the two contrasts: pre-vs. during-stim (left) and pre-vs. post-stim (right). [↗](#)

Figure S6. Analysis of pseudo-population activity within regions, separated by laterality relative to stimulation. Pseudo-population activity was characterized within each region via a linear dimensionality reduction on the trial-averaged firing rates. The temporal dynamics of each region's first three principal components are shown for (A) units ipsilateral to stimulation and (B) units contralateral to stimulation. HIP = hippocampus (coral), OFC = orbitofrontal cortex (yellow), AMY = amygdala (blue), ACC = anterior cingulate cortex (purple). [↗](#)

Figure S7. Control analyses for the detection of modulated units. The same permutation-based analyses reported in the manuscript were repeated under different control conditions. The percent of units (total $n = 203$) that met the firing rate threshold for inclusion (pink), the percent of included units modulated in the stim condition (purple), and the percent of included units modulated in the no-stim condition are shown. (A) The threshold for inclusion of units was varied from 0–3 Hz (0.1 step size); the black dashed line represents the ≥ 0.1 Hz threshold used in the manuscript. (B) To control for the possibility that non-physiological stimulation artifacts may preclude the detection of temporally adjacent spiking, we removed segments of data beginning at the onset of each burst of pulses (0–60 ms, 5 ms step size). Identical temporal windows were removed from the corresponding pre- and post-stimulation epochs to mitigate effects resulting solely from summation over different epoch sizes (reduced spike counts with shorter windows). (C) Visualization of the predicted probability of detecting modulation across synthetic neurons with variable firing rates and modulation effect sizes; FR = firing rate. [↗](#)

Figure S8. Analysis of multiunit activity response to stimulation. (A) Example trace of multiunit activity (MUA) in one channel during a single stimulation trial. Threshold crossings are highlighted with a pink dot overlaid on the MUA signal with a corresponding hash below. (B) The percentage of channels with significantly modulated MUA, separated by the direction of effect. (C) The percentage of channels with significantly modulated MUA, separated by direction effect and region. Inc (red; post > pre) vs. Dec (blue; post < pre). HIP = hippocampus, OFC = orbitofrontal cortex, AMY = amygdala, ACC = anterior cingulate cortex. *** $p < 0.001$, NS = not significant. [↗](#)

Figure S9. The effect of stimulation proximity to white matter and distance to recorded neurons. (A) Kernel density estimate of the Euclidean distance from stimulation contacts to nearest WM structure (in mm); hash marks represent individual observations. (B) The change in memory performance ($\Delta d'$) was linearly regressed onto the distance from the stimulated contacts to white matter. [↗](#)

Table S1 [↗](#)

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Reviewer #1 (Public review):

In this manuscript, Campbell et al. assess how intracranial theta-burst stimulation (TBS) applied to the basolateral amygdala in 23 epilepsy patients affects neuronal spiking in the medial temporal lobe and prefrontal cortex during a visual recognition memory task. This is an incredibly rare dataset; collecting single-unit spiking data from behaving humans during active intracranial stimulation is a Herculean task, with immense potential for translational studies of how stimulation may be applied to modulate biological mechanisms of memory. The authors utilize careful, high quality methodology throughout (e.g. task design, spike recording and sorting, statistical analysis), providing high confidence in the validity of their findings.

In providing such a detailed and deep investigation into the single-unit responses to intracranial stimulation the authors provide a very useful resources to any researchers in the fields of brain stimulation and human neurophysiology. This work could be instrumental in guiding diverse research studies, from basic science investigating the role of theta oscillations in human cognition to translational work investigating deep-brain stimulation for memory.

The authors have adequately addressed all prior concerns.

<https://doi.org/10.7554/eLife.106481.2.sa2>

Reviewer #2 (Public review):

Summary:

This study presents a valuable characterization of the effects of intracranial theta-burst stimulation of the basolateral amygdala on single units spiking activity in several areas in the human brain, associated with memory processing. It is written clearly and concisely, allowing readers to fully understand the analysis used.

The authors used a visual recognition memory task previously employed by their group to characterize the effects of basolateral amygdala stimulation upon memory consolidation

(Inman et al, 2018). This current report presents an interesting analysis that complements the results reported in the 2018 paper.

Strengths:

Rare combination of human neurophysiology and behavior -

The type of experiment performed in the manuscript, which contains both neurophysiological data, behavior, and a deep brain stimulation intervention (DBS), is incredibly rare, takes many years to accomplish with tight collaboration between clinical and research teams. Our understanding of spiking dynamics of human neurons is very limited, and this report is an important piece in the puzzle that allows DBS to be used in future interventions that will benefit patients' health.

Multiple brain areas included -

It's important to note that the report analyzes brain areas with which the Amygdala has extensive connections (Fig. 1A) - Hippocampus, OFC, Amygdala, ACC. It seems that neurons in all these areas were modulated by the stimulation, except the ACC, in which firing rates were so low that only a handful of neurons were included in the analysis. This is an important demonstration that low-amplitude stimulation (even when reduced to 0.5mA) can travel far and wide across the human brain.

The experiment is cleverly designed to tease apart responses due to visual stimuli (image presentation) and electrical stimulation. Authors suggest that the units modulated by stimulation are largely distinct from those responsive to image offset during trials without stimulation. The subpopulation that responds strongly also tends to have a higher baseline firing rate. It's important to add that the chosen modulation index is more likely to be significant in neurons with higher firing rates (Figure S8). The authors discuss the tradeoff of using a nonparametric modulation index for vs. other methods (for example, percent change in trial-averaged firing rate from baseline).

<https://doi.org/10.7554/eLife.106481.2.sa1>

Author response:

The following is the authors' response to the original reviews.

Reviewer #1 (Public review):

"This is an exploratory study that doesn't explore quite enough. Critically, the authors make a point of mentioning that neuronal firing properties vary across cell types, but only use baseline firing rate as a proxy metric for cell type. This leaves several important explorations on the table, not limited to the following:"

1a: "Do waveform shape features, which can also be informative of cell type, predict the effect of stimulation?"

To address this question, we modeled our approach to cell type classification after Peyrache et al. 2012. More specifically, we extracted two features from the mean unit waveforms—the valley-to-peak time (VP) and the peak half-width (PHW). These features were then used to classify units into two distinct clusters (k-means, clusters = 2, based on a strong prior from existing literature), representing putative excitatory and inhibitory neurons. Our approach recapitulated many of the same observations in Peyrache et al. 2012, namely (1) identification of two clusters (low PHW/VP: inhibitory, high PHW/VP: excitatory), (2) an ~80/20 ratio of excitatory/inhibitory neurons, and (3) greater baseline firing rates in the inhibitory vs. excitatory neurons. However, we did not observe a preferential modulation of one cell type

compared to another (see newly created Figure 4). A description of this analysis and its takeaways has been incorporated into the manuscript.

Change to Text:

Created Figure 4 (Separation of presumed excitatory and inhibitory neurons by waveform morphology).

Caption: (A) Two metrics were calculated using the averaged waveforms for each detected unit: the valley-to-peak width (VP) and peak half-width (PHW). (B) Scatterplot of the relationship between VP and PHW; note that units with identical metrics are overlaid. Using k-means clustering, we identified two distinct response clusters, representing presumed excitatory (E, blue) and inhibitory (I, red) neurons. The units from which the example waveforms were taken are outlined in black. Probability distributions for each metric are shown along the axes. (C) Total number of units within each cluster, separated by region. (D) Comparison of baseline firing rates, separated by cluster. (E) Percent of modulated units in each cluster. * $p < 0.05$, NS = not significant.

Added a description of clustering methodology to lines 132-137: “We calculated two metrics from the averaged waveform from each detected unit: the valley-to-peak-width (VP) and the peak half-width (PHW) (Figure 4A); previously, these two properties of waveform morphology have been used to discriminate pyramidal cells (excitatory) from interneurons (inhibitory) in human intracranial recordings (Peyrache et al., 2012). Next, we performed k-means clustering ($n = 2$ clusters) on the waveform metrics, in line with previous approaches to cell type classification.

Added a section in the Results titled “Theta Burst Stimulation Modulates Excitatory and Inhibitory Neurons Equally”. Lines 370-378: “Using k-means clustering, we grouped neurons into two distinct clusters based on waveform morphology, representing neurons that were presumed to be excitatory (E) and inhibitory (I) (Figure 4B). Inhibitory (fast-spiking) neurons exhibited shorter waveform VP and PHW, compared with excitatory (regular-spiking) neurons (I cluster centroid: VP = 0.50ms, PHW = 0.51ms; E cluster centroid: VP = 0.32ms, PHW = 0.31ms), and greater baseline firing rates ($U(N_I = 23, N_E = 133) = 1074.50, p = 0.023$) (Figure 4D). Although we observed a much greater proportion of excitatory vs. inhibitory neurons (E: 85.3%, I: 14.7%), stimulation appeared to affect excitatory and inhibitory neurons equally, suggesting that one cell type is not preferentially activated over another (Figure 4E).

Modified discussion of the effects of stimulation on different cell types. Lines 475-483: “...To test these hypotheses directly, we clustered neurons into presumed excitatory and inhibitory neurons based on waveform morphology. In doing so, we observed ~85% excitatory and ~15% inhibitory neurons, which is very similar what has been reported previously in human intracranial recordings (Cowan et al. 2024, Peyrache et al., 2012). Interestingly, stimulation appeared to modulate approximately the same proportion of neurons for each cell type (~30%), despite the differently-sized groups. Recent reports, however, have suggested that the extent to which electrical fields entrain neuronal spiking, particularly with respect to phase-locking, may be specific to distinct classes of cells (Lee et al., 2024).”

1b: “Is the autocorrelation of spike timing, which can be informative about temporal dynamics, altered by stimulation? This is especially interesting if theta-burst stimulation either entrains theta-rhythmic spiking or is more modulatory of endogenously theta-modulated units.”

The reviewer is correct in suggesting that rate-modulation represents only one of many possible ways by which exogenous theta burst stimulation may influence neuronal activity. Indeed, intracranial theta burst stimulation has previously been shown to evoke theta-frequency oscillatory responses in local field potentials (Solomon et al. 2021), and other forms

of stimulation (i.e., transcranial alternating current stimulation) may modulate the rhythm, rather than the rate, of neuronal spiking (Krause et al. 2019).

To investigate whether stimulation altered rhythmicity in neuronal firing, we contrasted the spike timing autocorrelograms, as suggested. More specifically, we computed the pairwise differences in spike timing for each trial, separating spikes into the same pre-, during-, and post-stimulation epochs described in the manuscript (bin size = 5 ms, max lag = 250 ms), grouped neurons by whether they were modulated, and then contrasted the differences in the latencies of the peak normalized autocorrelation value between epochs. Only neurons with a firing rate of ≥ 1 Hz ($n = 70/203$, 34.5%) were included in this analysis since sparse firing resulted in noisy autocorrelation estimates. Subsequent statistical testing of the peak latency differences between pre-/during- and pre-/post-stimulation did not reveal any group-level differences (Mann-Whitney U tests, $p > 0.05$). Thus, we were not able to identify neuronal responses suggestive of altered rhythmicity (see Figure S5). A description of this analysis and its takeaways has been incorporated into the manuscript.

Of note, there are two elements of the data that constrain our ability to detect modulation in the rhythm of firing. First, the baseline activity recorded across neurons modulated by stimulation was relatively low (i.e., median firing rate = 1.77 Hz). Second, stimulation often resulted in a suppression, rather than an enhancement, of firing rate. Taken together, the sparse firing afforded limited opportunity to characterize changes to subtle patterns of spiking.

Change to Text:

Created Figure S5 (Analysis of modulation in spiking rhythmicity)

Caption: (A) Representative autocorrelograms (ACG) for a single neuron. The pairwise differences in spike timing were computed for each trial and epoch (bin size = 5 ms, max lag = 250 ms), then smoothed with a Gaussian kernel. The peak in the normalized ACG across trials was computed for each epoch. (B) Kernel density estimate of the peak ACG lag, separated by epoch. (C) The peak ACG lags were split by whether the neuron was modulated (Mod) or unaffected by stimulation (NS = not significant) for each of the two contrasts: pre- vs. during-stim (left) and pre- vs. post-stim (right).

Details about the autocorrelation methodology have been incorporated. Lines 166-172: “To investigate whether stimulation altered rhythmicity in neuronal firing, we analyzed the spike timing autocorrelograms. More specifically, we computed the pairwise differences in spike timing for each trial (bin size = 5 ms, max lag = 250 ms) and then contrasted the differences in the latencies of the peak normalized autocorrelation value between epochs (pre-, during-, post-stimulation). Only neurons with a firing rate of ≥ 1 Hz ($n = 70/203$, 34.5%) were included in this analysis since sparse firing resulted in noisy autocorrelation estimates.

The results from contrasting the autocorrelograms are now mentioned briefly. Lines 297-298: “Stimulation, however, did not appear to alter the rhythmicity in neuronal firing, as measured by spiking autocorrelograms (Figure S5).”

1c: “The authors reference the relevance of spike-field synchrony (30-55 Hz) in animal work, but ignore it here. Does spike-field synchrony (comparing the image presentation to post-stimulation) change in this frequency range? This does not seem beyond the scope of investigation here.”

We agree that a further characterization of spike-field and spike-phase relationships may provide rich insights into more complex regional and interregional dynamics that may be altered by stimulation. Given that many metrics are biased by sample size (e.g., number of spikes), which can vary considerably, computing the pairwise phase consistency (PPC)

between spikes and LFP is a preferred metric (Vinck et al. 2010). Although PPC is unbiased, its variance nonetheless increases considerably with low spike counts; pooling spike counts across trials, however, decouples the temporal relationship between spiking and the LFP phase for each trial, confounding results and yielding an unstable estimate.

To determine whether such an analysis is indeed possible, we calculated the percentage of stimulation trials with ≥ 10 spikes in both the 1s pre- and post-stimulation epochs (a relatively low threshold for inclusion). Only a very small proportion of the total number of trials across all neurons met this criterion (2.5%). Thus, because of the sparse spiking in our data, we are unable to reliably characterize spike-field or spike-phase modulation in detected neurons.

Change to Text:

In the manuscript, we have added a description of why our data is not well-suited to investigate these relationships.

Lines 532-538: “The present study did not investigate interactions between spiking activity and local field potentials because neuronal spiking was sparse at baseline and often further suppressed by stimulation; only a very small proportion of the total number of trials across all neurons exhibited ≥ 10 spikes in both the 1s pre- and post-stimulation epochs (~2.5%). Although certain metrics are not biased by sample size (e.g., pairwise phase consistency), low spike counts can dramatically affect variance and, therefore, result in unstable estimates (Vinck et al., 2011).

1d: “How does multi-unit activity respond to stimulation? At this somewhat low count of neurons (total n=156 included) it would be valuable to provide input on multi-unit responses to stimulation as well.”

We thank the reviewer for this suggestion. We have incorporated an analysis of multiunit activity (MUA), which similarly identifies robust modulation via permutation-based statistical testing and characterizes the different profiles of responses (i.e., increased vs. decreased MUA threshold crossings pre- vs. post-stimulation).

Change to Text:

Created Figure S8 (Analysis of multiunit activity response to stimulation)

Caption: (A) Example trace of multiunit activity (MUA) in one channel during a single stimulation trial. Threshold crossings are highlighted with a pink dot overlaid on the MUA signal with a corresponding hash below. (B) The percentage of channels with significantly modulated MUA, separated by the direction of effect. (C) The percentage of channels with significantly modulated MUA, separated by direction effect and region. Inc (red; post > pre) vs. Dec (blue; post < pre). HIP = hippocampus, OFC = orbitofrontal cortex, AMY = amygdala, ACC = anterior cingulate cortex. *** $p < 0.001$, NS = not significant.

Details about the MUA methodology have been incorporated. Lines 174-180: “Finally, we measured modulation in multiunit activity (MUA) by filtering the microelectrode signals in a 300-3,000 Hz window and counting the number of threshold crossings. Thresholds were determined on a per-channel basis and defined as -3.5 times the root mean square of the signal during the baseline period; activity during stimulation was excluded since stimulation artifact is difficult to separate from MUA in the absence of spike sorting.

MUA results are now incorporated. Lines 365-367: “Additional characterization of MUA revealed a dominant signature of increased activity post- vs. pre-stimulation, in line with these trends observed at the single-neuron level (Figure S8).”

1e: "Several intracranial studies have implicated proximity to white matter in determining the effects of stimulation on LFPs; do the authors see an effect of white matter proximity here?"

We thank the reviewer for the interesting question. Subsequent characterization revealed only small differences in the proximity of stimulation contacts to white matter (range 1.5-8.0 mm), likely because the chosen target (i.e., basolateral amygdala) has several nearby white matter structures (e.g., stria terminalis). Nonetheless, we performed a linear regression between the proximity to white matter and the stimulation-induced effect on behavior (stimulation vs. no-stimulation d' difference), the results of which indicate no clear association ($p > 0.05$; see Figure S9). Critically, this is not to suggest that white matter proximity has no interaction with the reported behavioral effects, but rather, that we could not identify such an association within our data.

Change to Text:

Created Figure S9 (The effect of stimulation proximity to white matter and distance to recorded neurons).

Caption: (A) Kernel density estimate of the Euclidean distance from stimulation contacts to nearest WM structure (in mm); hash marks represent individual observations. (B) The change in memory performance ($\Delta d'$) was linearly regressed onto the distance from the stimulated contacts to white matter.

The following has been added to lines 405-426: "Proximity to white matter has been shown to influence the effects of stimulation on behavior and the strength of evoked responses (Mankin et al., 2021; Mohan et al., 2020; Paulk et al., 2022). Across all stimulated contacts, we observed only small differences in the proximity of stimulation contacts to white matter (median = 4.5 mm, range = 1.5-8.0 mm), likely because the chosen target (i.e., basolateral amygdala) has several nearby white matter structures (e.g., stria terminalis). Nonetheless, we performed a linear regression between the proximity to white matter and the stimulation-induced effect on behavior (stimulation vs. no-stimulation d' difference), the results of which indicate no clear association ($p > 0.05$; see Figure S9).

Comment 2: "It is a little confusing to interpret stimulation-induced modulation of neuronal spiking in the absence of stimulation-induced change in behavior. How do the authors findings tell us anything about the neural mechanisms of stimulation-modulated memory if memory isn't altered? In line with point #1, I would suggest a deeper dive into behavior (e.g. reaction time? Or focus on individual sessions that do change in Figure 4A?) to make a stronger statement connecting the neural results to behavioral relevance."

We agree that the connection between the observed stimulation-induced neuronal modulation and effects on behavior is unclear and has proven challenging to elucidate. Per the reviewer's suggestion, we further focused our analyses on the neuronal modulation effects in the individual sessions that resulted in a robust change in memory performance (stimulation vs. no-stimulation d' difference threshold of ± 0.5 , based on a moderate effect size for Cohen's d); both a positive and negative threshold were used to capture robust changes in memory performance associated with firing rate modulation, whether enhancement or suppression. To this end, we contrasted the proportion of modulated neurons in the sessions where stimulation resulted in a robust behavioral change ($\Delta d'$) with those that did not ($\sim d'$). We did not observe a difference in the proportions between groups when collapsed across all sampled regions, or when separately evaluated (Fisher's exact tests, $p > 0.05$; see Figure 5C).

Given that this approach did not further clarify the connection between our neural and behavioral results, we believe it is most appropriate to deemphasize claims in the manuscript regarding the potential insights for behavioral modulation (e.g., memory enhancement), and have done so.

Change to Text:

Toned down reference to the memory-related effects of stimulation in the abstract by removing the following lines from the abstract: “Previously, we demonstrated that intracranial theta burst stimulation (TBS) of the basolateral amygdala (BLA) can enhance declarative memory, likely by modulating hippocampal-dependent memory consolidation...” and “...and motivate future neuromodulatory therapies that aim to recapitulate specific patterns of activity implicated in cognition and memory.”

Changed Figure 4 to Figure 5

Created Figure 5C (Interaction between behavioral effects and neuronal modulation)(C) Change in recognition memory performance was split into two categories using a d' difference threshold of ± 0.5 : responder (positive or negative; $\Delta d'$, pink) and non-responder ($\sim d'$, grey). Individual d' scores are shown (left) with points colored by outcome category; dotted lines demarcate category boundaries, and the grey-shaded region represents negligible change. The number of sessions within each outcome category (middle) and the proportion of modulated units as a function of outcome category, separated by region (right). NS = not significant.

The description of the behavioral results has been updated. Lines 394-403: “At the level of individual sessions, we observed enhanced memory ($\Delta d' > +0.5$) in 36.7%, impaired memory ($\Delta d' < -0.5$) in 20.0%, and negligible change ($-0.5 \leq \Delta d' \leq 0.5$) in 43.3% when comparing performance between the stim and no-stim conditions; a threshold of $\Delta d' \pm 0.5$ was chosen for this classification based on the defined range of a “medium effect” for Cohen’s d . To test our hypothesis that neuronal modulation would be associated with changes in memory performance, we combined the sessions that resulted in either memory enhancement or impairment and contrasted the proportion of modulated units across regions sampled. We did not, however, observe a meaningful difference in the proportion of modulated units when grouped by behavioral outcome (all contrasts $p > 0.05$) (Figure 5C).

Lines 213-214 and 394-397 have been edited to reflect a change in the d' threshold used for categorizing behavioral results (from $\Delta d' \pm 0.2$ to $\Delta d' \pm 0.5$).

Comment 3: “It is not clear to me why the assessment of firing rates after image onset and after stim offset is limited to one second - this choice should be more theoretically justified, particularly for regions that spike as sparsely as these.”

We thank the reviewer for this question and acknowledge that no clear justification was provided for this decision in the manuscript. Our decision to limit each of the analysis epochs to 1s was chosen for two reasons. First, the maximum possible length of the during-stimulation epoch was 1 s (stim on for 1 s). Although the pre- and post-stimulation epochs could be extended without issue, we were concerned that variable time windows could introduce a bias, for instance, resulting in different variances between epochs. Second, we anticipated, both from empirical observations and prior literature, that the neural response following stimulation or task features (e.g., image onset/offset) was likely to be transient, rather than sustained for a period of many seconds. By keeping the windows short, we ensured that our approach to detecting modulation (i.e., contrasting trial-wise spike counts between each pair of epochs) captured the intended effect rather than random noise. We

have incorporated a discussion of this rationale in the Peri-Stimulation Modulation Analyses section.

Change to Text:

Lines 156-158 have been added: “Each epoch was constrained to 1 s to ensure that subsequent firing rate contrasts were unbiased and to capture potential transient effects (e.g., image onset/offset).”

Comment 4: “This work coincides with another example of human intracranial stimulation investigating the effect on firing rates (doi: <https://doi.org/10.1101/2024.11.28.625915>). Given how incredibly rare this type of work is, I think the authors should discuss how their work converges with this work (or doesn't).”

Thank you for bringing this highly relevant work to our attention. We were unaware of this recent preprint and have incorporated a discussion of its main findings into the manuscript.

Change to Text:

New citations: van der Plas et al. 2024 (bioRxiv), Cowan et al. 2024 (bioRxiv)

The discussion of related studies has been updated. Lines 447-457: “Few studies, however, have characterized the impact of electrical stimulation via macroelectrodes on the spiking activity of human cortical neurons, none of which involve intracranial theta burst stimulation. One study reported a long-lasting reduction in neural excitability among parietal neurons, with variable onset time and recovery following continuous transcranial TBS in non-human primates (Romero et al., 2022). In a similar vein, it was recently shown that human neurons are largely suppressed by single-pulse electrical stimulation (Cowan et al., 2024; Plas et al., 2024). Other emerging evidence suggests that transcranial direct current stimulation may entrain the rhythm rather than rate of neuronal spiking (Krause et al., 2019) and that stimulation-evoked modulation of spiking may meaningfully impact behavioral performance on cognitive tasks (Fehring et al., 2024).”

Comment 5: “What information does the pseudo-population analysis add? It's not totally clear to me.”

We recognize the need to further contextualize the motivation for the exploratory pseudo-population analysis and appreciate the reviewer for bringing the lack of detail to our attention. In brief, the analysis allowed us to observe trends in activity across populations of neurons, which, in principle, are not visible by characterizing modulation solely in discrete neurons. Additional details have been incorporated into the manuscript, as suggested.

Change to Text:

Additional justification has been incorporated in the description of the methodology. Lines 185-187: “...This approach enables the identification of dominant patterns of coordinated neural activity that may not be apparent when examining individual neurons in isolation.”, lines 192-194: “...By collapsing across subjects into a common pseudo-population, this analysis provides a mesoscale view of how stimulation modulates shared activity patterns across anatomically distributed neural populations.”

A summary interpretation has been added to the paragraph describing the results. Lines 326-328: “Taken together, these analyses reveal global structure in the state space of responses to BLA stimulation within hippocampal circuits.”

Reviewer #2(Public review):

Comment 1 "Authors suggest that the units modulated by stimulation are largely distinct from those responsive to image offset during trials without stimulation. The subpopulation that responds strongly also tends to have a higher baseline of firing rate. It's important to add that the chosen modulation index is more likely to be significant in neurons with higher firing rates."

This is an important point that was not previously addressed in our manuscript. We suspect there are likely two factors at play worth considering with respect to our chosen nonparametric modulation index: neurons with lower activity require smaller changes in spike counts to be significantly modulated (easier to flip ranks), and neurons with higher activity empirically exhibit greater absolute shifts in the number of spikes. Our further use of permutation testing, while mitigating false positives, may also somewhat constrain the ability to detect modulation in sparsely active neurons. Nonetheless, given that many trials entailed few or no spikes, we believe this approach is preferable to alternatives that may be more susceptible to noise (e.g., percent change in trial-averaged firing rate from baseline).

To better understand the tradeoffs with detection probability, we performed a sensitivity analysis. We generated synthetic data with different baseline firing rates (0.1-5.0 Hz) and effect sizes (± 0.1 -0.7 Hz) and simulated the likelihood of detection with our given modulation index across neurons. The results of the simulation support the notion that the probability of detecting modulation is lower for sparsely active neurons (Figure S8C). Further discussion of this consideration for the chosen modulation index, as well as details regarding the sensitivity analysis, have been incorporated into the manuscript.

Change to Text:

Created Figure S7C (Detection probability analysis)

Caption: The same permutation-based analyses reported in the manuscript were repeated under different control conditions... (C) Visualization of the predicted probability of detecting modulation across synthetic neurons with variable firing rates and modulation effect sizes; FR = firing rate.

Lines 223-224 have been added to the Methods section titled "Firing Rate Control Analyses": "We performed a series of control analyses to test whether our approach to firing rate detection was robust..."

A description of the simulation has been incorporated into the same section as above. Lines 234-237: "Finally, to better understand the tradeoffs with our statistical approach, we generated synthetic data with different baseline firing rates (0.1-5.0 Hz) and effect sizes (± 0.1 -0.7 Hz), then simulated the likelihood of detecting modulation across variable conditions (Figure S7C)."

The description of the results from the control analyses has been updated. Lines 330-339: "Finally, we performed three supplementary analyses to evaluate the robustness of our approach to detecting firing rate modulation: a sensitivity analysis assessing the proportion of modulated units at different firing rate thresholds for inclusion/exclusion, a data dropout analysis designed to control for the possibility that non-physiological stimulation artifacts may preclude the detection of temporally adjacent spiking, and a synthetic detection probability analysis. These results recapitulate our observation that units with higher baseline firing are most likely to exhibit modulation (though the probability of detecting modulation is lower for sparsely active neurons) and suggest that suppression in firing rate is not solely attributable to amplifier saturation following stimulation (Figure S7).

Comment 2: "Readers can benefit from understanding with more details the locations chosen for stimulation - in light of previous studies that found differences between effects based on proximity to white matter (For example - PMID 32446925, Mohan et al, Brain Stimul. 2020 and PMID 33279717 Mankin et al Brain Stimul. 2021)."

This has been addressed in the above response to Reviewer's 1 comment 1.1e.

Change to Text:

See changes related to Reviewer 1 comment 1.1e.

Comment 3: "Missing information in the manuscript..."

3a: "Images of stimulation anatomical locations for all subjects included in this study. Ideally information about the impedance of the contacts to be able to calculate the actual current used."

As requested, we have provided an image from the coronal T1 MRI sequence, which highlights the position of the stimulated contacts for each of the 16 patients. Though we did not measure the impedances directly, the stimulation was current-controlled, which ensured that the desired current and charge density were consistent regardless of the tissue or electrode impedance.

Change to Text:

Created Figure S1 (Anatomical location of stimulated electrodes).

Caption: A coronal slice from the T1-weighted MRI scan is shown for each patient who participated in the study (n = 16). Electrode contacts within the same plane of the image are shown with blue circles, and the bipolar pair of stimulated contacts within the basolateral amygdala is highlighted in red.

Lines 144-145 have been edited to reflect that the delivered stimulation was current-controlled: "Specifically, we administered current-controlled, charge-balanced, ..."

3b: "The studied population is epilepsy patients, and the manuscript lacks description of their condition, proximity to electrodes included in the study to pathological areas, and the number of units from each patient/hemisphere."

We agree that additional information regarding patient demographics, experimental details, and clinical characteristics would further contextualize this unique patient population. A new table has been included, which contains the following information: patient ID, sex, age, # experimental session, # SEEG leads (and # microelectrodes), # detected units (L vs. R hemisphere), and suspected seizure onset zone.

Change to Text:

Created Table S1 (Patient demographics and clinical characteristics).

Lines 258-259 have been added: "... (see Table S1 for patient demographics)."

3c: "I haven't seen any comments on code availability (calculating modulation indices and statistics) and data sharing."

For clarification, a section titled Resource Availability is already appended to the end of the manuscript following the Conclusion, which describes the data and code availability.

Change to Text:

None

| 3d: *"Small comment - Figure legend 3E - Define gray markers (non-modulated units?)"*

Thank you for highlighting this omission. We have updated the relevant figure caption.

Change to Text:

The following has been added to the Figure 3 caption: "...whereas units without a significant change in activity are shown in grey."

<https://doi.org/10.7554/eLife.106481.2.sa0>